Power BI Snooker Analysis Project

```
PlayerID,PlayerName,Country ,Ranking ,Age
1,Edwards,England,4,31
2,Kimberson,Germany,2,19
3,Longen,China,5,24
4,Johnson-Smith-Peters,Belgium,1,26
5,Kylie,Poland,3,29
6,Samuels,US,6,30
7,Mitchem,Canada,12,17
8,Doe,China,8,25
9,Stevensons-Hu,England,11,29
10,Nicholsons,England,10,22
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MatchID,Tournament,Season,Date,Player1,Player2,Winner,FramesWonP1,FramesWonP2,CenturyBreaks,HighestBreak,MatchDuration,Round,Venue
1,World League,2023/2024,02/10/2023,Edwards,Sebastian,Edwards,5,3,2,118,142,Round of 32,London Arena
2,World League,2023/2024,03/10/2023,Johnson-Smith-Peters,Edwardo,Edwardo,4,5,1,102,151,Round of 32,London Arena
3,World League,2023/2024,04/10/2023,Samuels,Neeman,Samuels,5,1,3,126,119,Round of 32,London Arena
4,World League,2023/2024,05/10/2023,Kylie,Carvalho,Kylie,5,4,2,109,168,Round of 32,London Arena
5,World League,2023/2024,06/10/2023,Edwards,Newton,Edwards,5,2,1,97,127,Round of 16,Manchester Central
6,World League,2023/2024,07/10/2023,Kylie,Willison,Kylie,5,3,2,111,135,Round of 16,Manchester Central
7,World League,2023/2024,08/10/2023,Johnson-Smith-Peters,Sebastian,Sebastian,3,5,1,101,144,Round of 16,Manchester Central
8,World League,2023/2024,09/10/2023,Kylie,Edwardo,Edwardo,4,5,3,133,176,Quarter Final,Manchester Central
9,China Championship,2024/2025,13/09/2024,Mitchem,Walters,Mitchem,5,2,2,105,137,Round of 32,Beijing Indoor Arena
10,China Championship,2024/2025,14/09/2024,Doe,Ferguson,Ferguson,2,5,1,99,128,Round of 32,Beijing Indoor Arena
11,China Championship,2024/2025,15/09/2024,Stevensons-Hu,Richardson,Stevensons-Hu,5,4,2,117,172,Round of 32,Beijing Indoor Arena
12,China Championship,2024/2025,16/09/2024,Longen,Mitchem,Longen,5,1,3,124,111,Round of 16,Shanghai Masters Hall
```
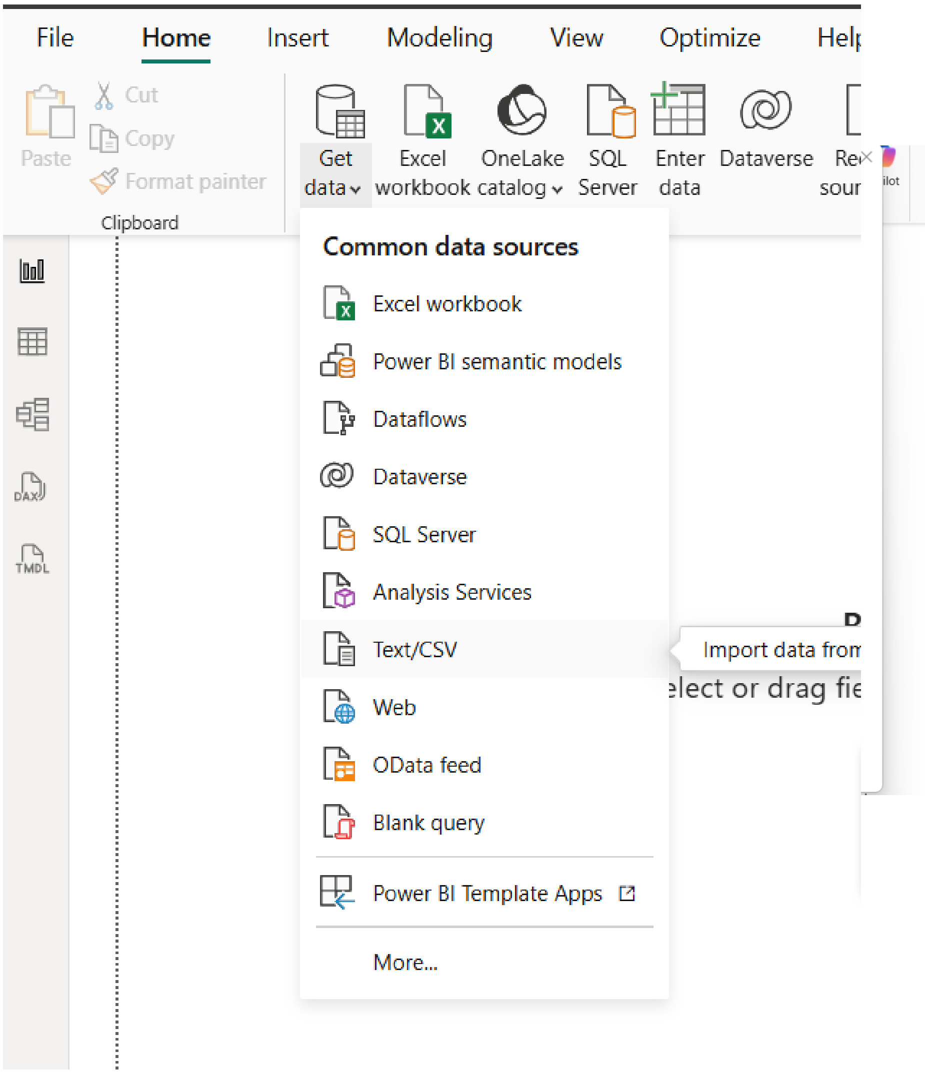

First we create a dataset of the information to analyse in PowerBI. Here we use 2 csv files to define our 2 tables to be imported into Excel these being a players table and a matches table, these screenshots show a subset of these two tables.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SQL Server
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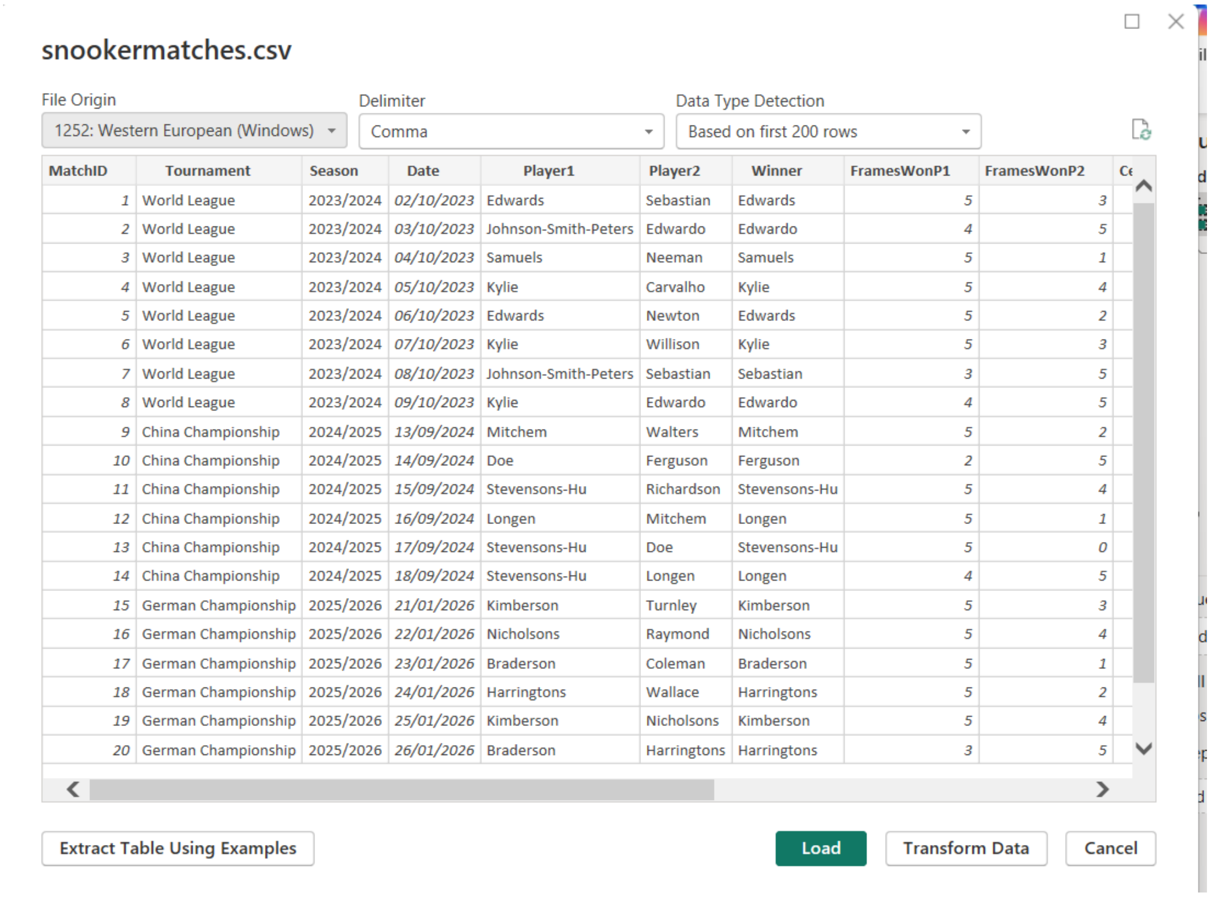
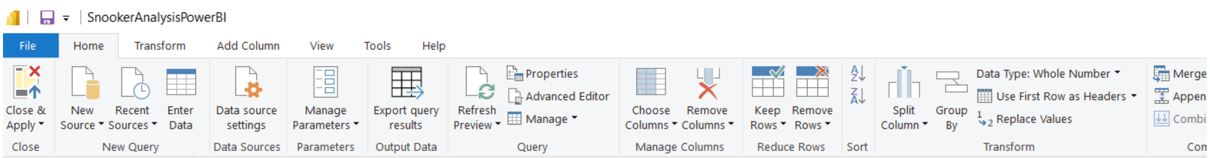
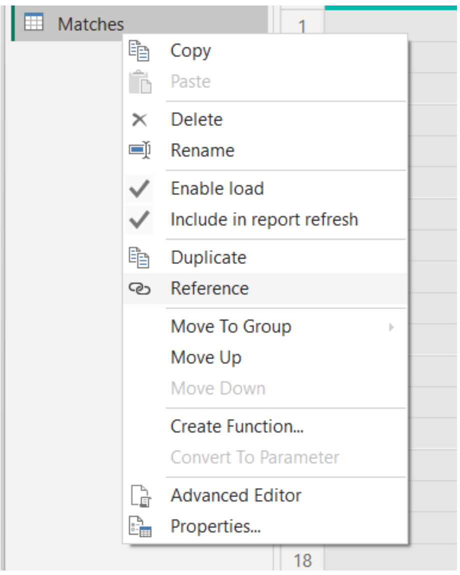
Dataverse
Common data sources
Excel workbook
Power BI semantic models
Dataflows
Dataverse
SQL Server
Analysis Services
Text/CSV
Web
OData feed
Blank query
Power BI Template Apps
More...
Import data from

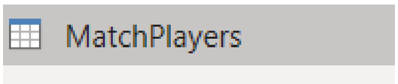


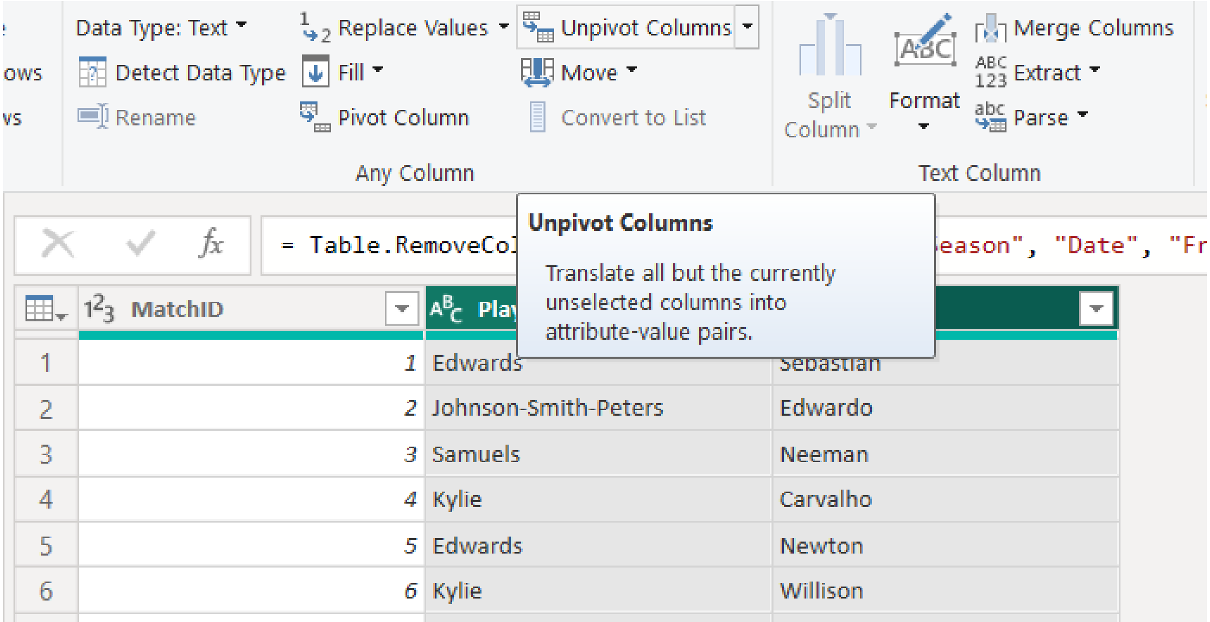
MatchID	Tournament	Season	Date	Player1	Player2	Winner	FramesWonP1	FramesWonP2	Ce
1	World League	2023/2024	02/10/2023	Edwards	Sebastian	Edwards	5	3	
2	World League	2023/2024	03/10/2023	Johnson-Smith-Peters	Edwardo	Edwardo	4	5	
3	World League	2023/2024	04/10/2023	Samuels	Neeman	Samuels	5	1	
4	World League	2023/2024	05/10/2023	Kylie	Carvalho	Kylie	5	4	
5	World League	2023/2024	06/10/2023	Edwards	Newton	Edwards	5	2	
6	World League	2023/2024	07/10/2023	Kylie	Willison	Kylie	5	3	
7	World League	2023/2024	08/10/2023	Johnson-Smith-Peters	Sebastian	Sebastian	3	5	
8	World League	2023/2024	09/10/2023	Kylie	Edwardo	Edwardo	4	5	
9	China Championship	2024/2025	13/09/2024	Mitchem	Walters	Mitchem	5	2	
10	China Championship	2024/2025	14/09/2024	Doe	Ferguson	Ferguson	2	5	
11	China Championship	2024/2025	15/09/2024	Stevensons-Hu	Richardson	Stevensons-Hu	5	4	
12	China Championship	2024/2025	16/09/2024	Longen	Mitchem	Longen	5	1	
13	China Championship	2024/2025	17/09/2024	Stevensons-Hu	Doe	Stevensons-Hu	5	0	
14	China Championship	2024/2025	18/09/2024	Stevensons-Hu	Longen	Longen	4	5	
15	German Championship	2025/2026	21/01/2026	Kimberson	Turnley	Kimberson	5	3	
16	German Championship	2025/2026	22/01/2026	Nicholsons	Raymond	Nicholsons	5	4	
17	German Championship	2025/2026	23/01/2026	Braderson	Coleman	Braderson	5	1	
18	German Championship	2025/2026	24/01/2026	Harringtons	Wallace	Harringtons	5	2	
19	German Championship	2025/2026	25/01/2026	Kimberson	Nicholsons	Kimberson	5	4	
20	German Championship	2025/2026	26/01/2026	Braderson	Harringtons	Harringtons	3	5	

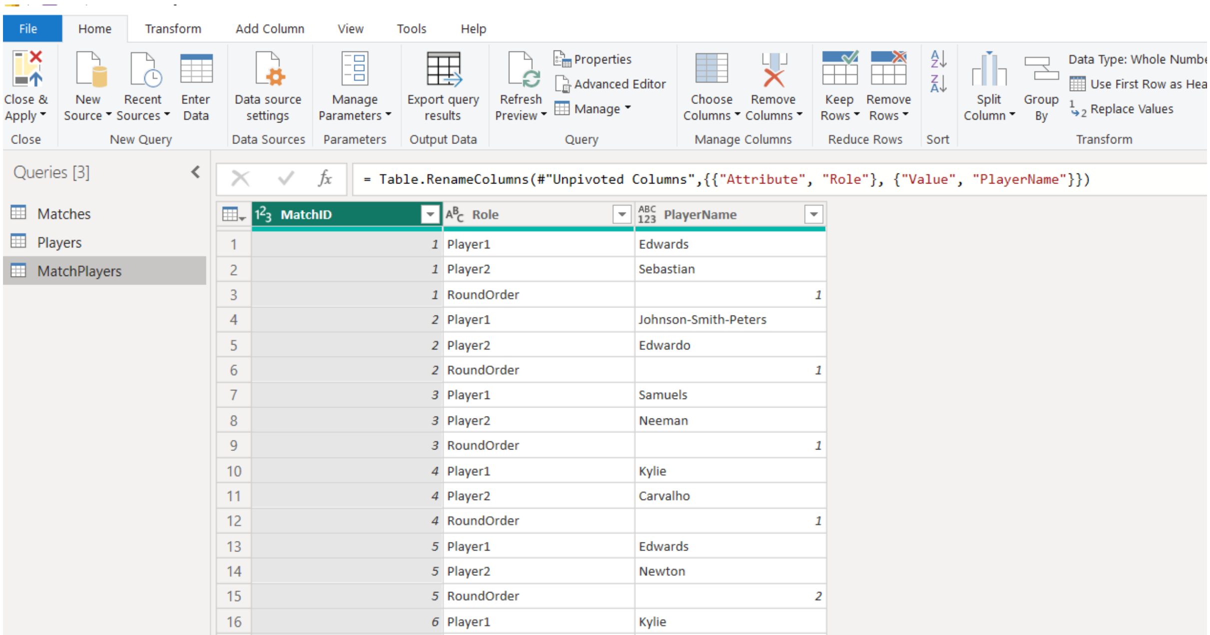
Next, in PowerBI we load in both the players.csv and the snookermatches.csv as our initial two tables to work with.







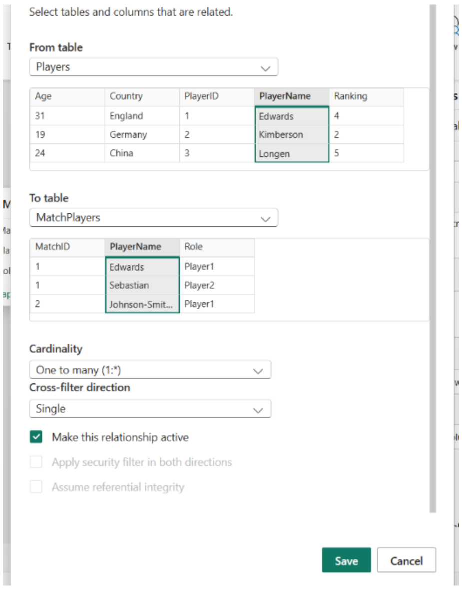




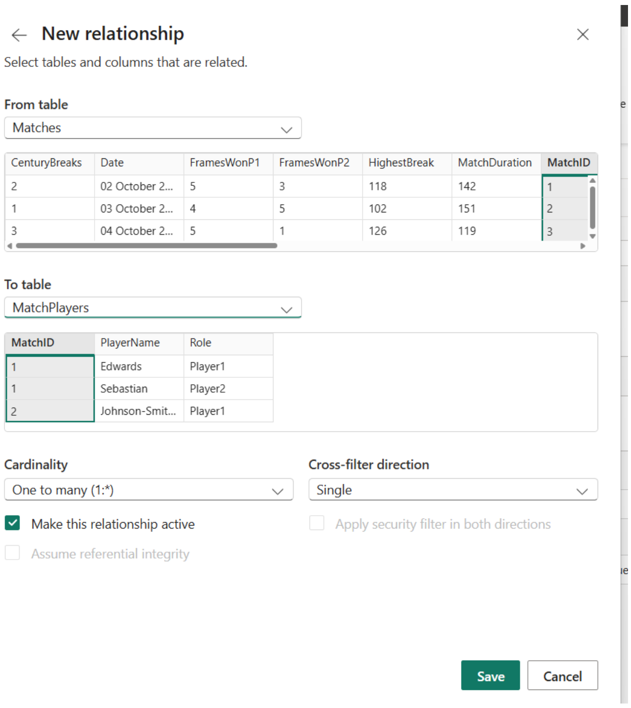
Here we in the power query editor we create a reference table from our Matches table and name this MatchPlayers. We then remove all columns from this table except MatchID, Player1 and Player2. Further to this we then select both Player1 and Player2 columns and unpivot these columns to create separate rows in our MatchPlayers table for each player contesting in a match.

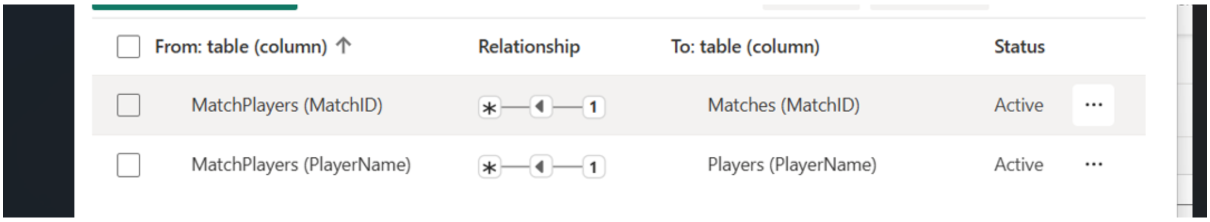
MatchID	Role	PlayerName
1	Player1	Edwards
1	Player2	Sebastian
2	Player1	Johnson-Smith-Peters
2	Player2	Edwardo
3	Player1	Samuels
3	Player2	Neeman
4	Player1	Kylie
4	Player2	Carvalho
5	Player1	Edwards
5	Player2	Newton
6	Player1	Kylie
6	Player2	Willison
7	Player1	Johnson-Smith-Peters
7	Player2	Sebastian
8	Player1	Kylie
8	Player2	Edwardo
9	Player1	Mitchem
9	Player2	Walters
10	Player1	Doe
10	Player2	Ferguson
11	Player1	Stevensons-Hu
11	Player2	Richardson

Both columns are then renamed from Attribute And Value to **Role and PlayerName**.



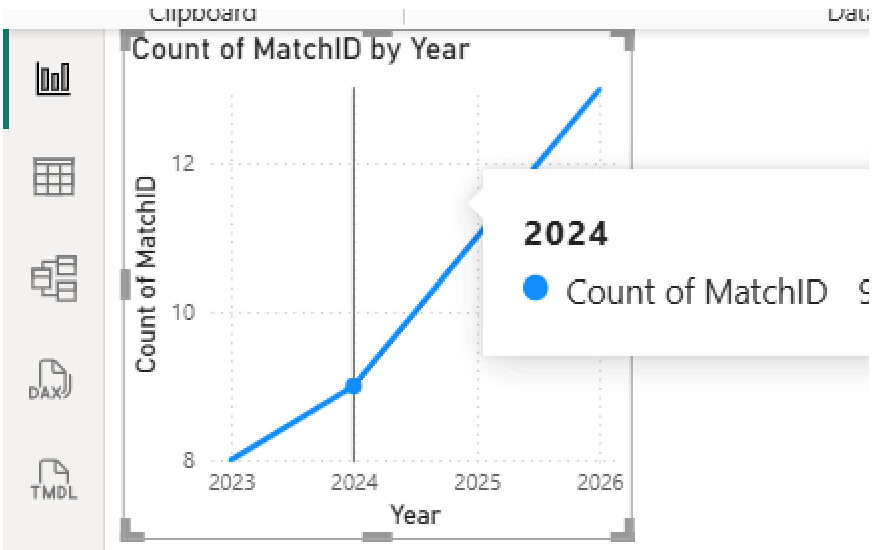
In the model view we define our relationships by right clicking a table such as Players. This will bring the following above screen where we define our relationship from the from Table to the to Table and we select the column that is the link between the 2 tables in this case it is PlayerName. We define a One To Many Relationship because a Player can have many MatchPlayer records but a MatchPlayer links to only one Player Record. We also define a One To Many Relationship between Matches And MatchPlayers with the column link this time being MatchID.

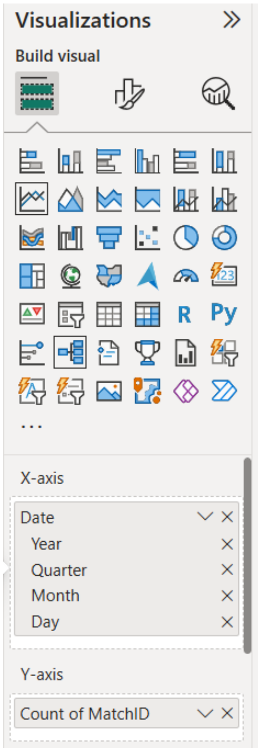




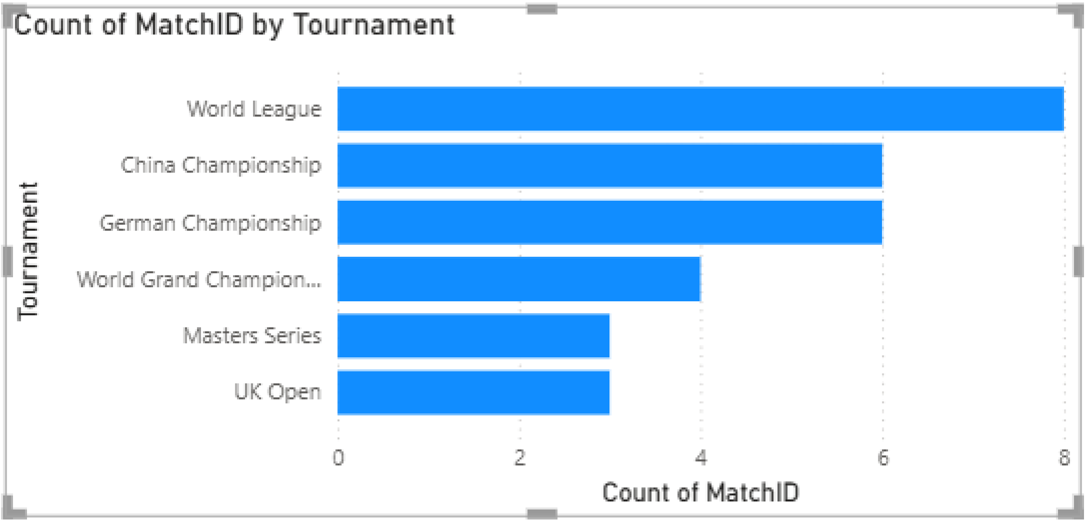
From: table (column)	Relationship	To: table (column)	Status
MatchPlayers (MatchID)	* — 1	Matches (MatchID)	Active
MatchPlayers (PlayerName)	* — 1	Players (PlayerName)	Active

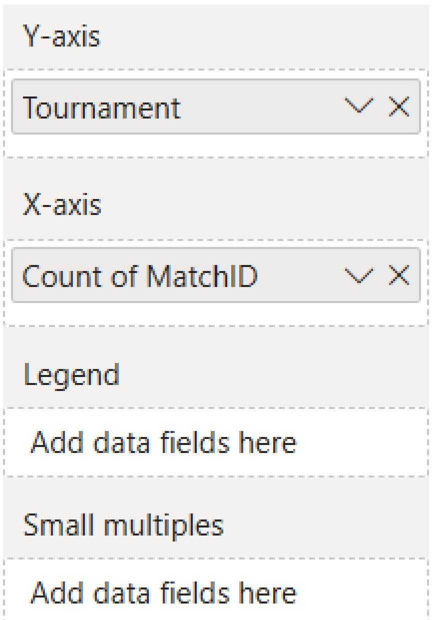
Now we are ready to start creating reports.



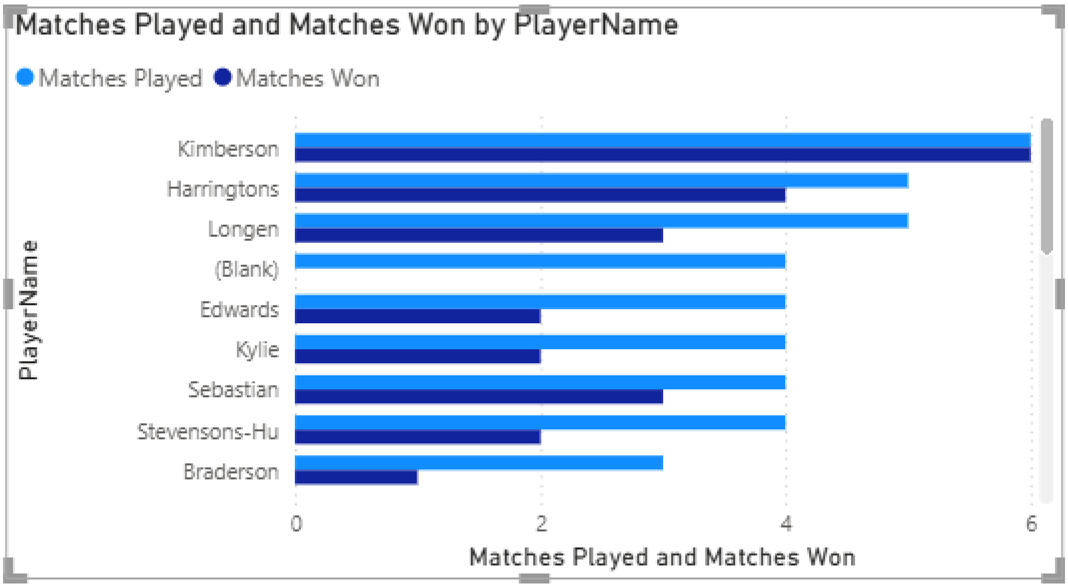


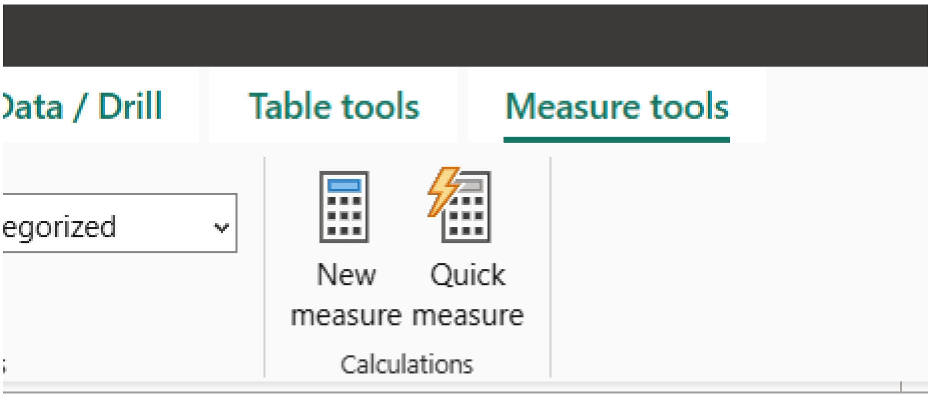
The first report created shows the total number of matches across the years as a line graph. To generate this we choose the date column from our matches table to represent x-axis and the count of match ids from our matches table.





The second report created shows the number of matches across the tournaments as a clustered bar chart. To generate this for the x-axis the Matches Tournament column is chosen and for the y-axis this is a count of the Matches Match Id's for each Tournament.






```
Matches Played =
DISTINCTCOUNT(MatchPlayers[MatchID])
```

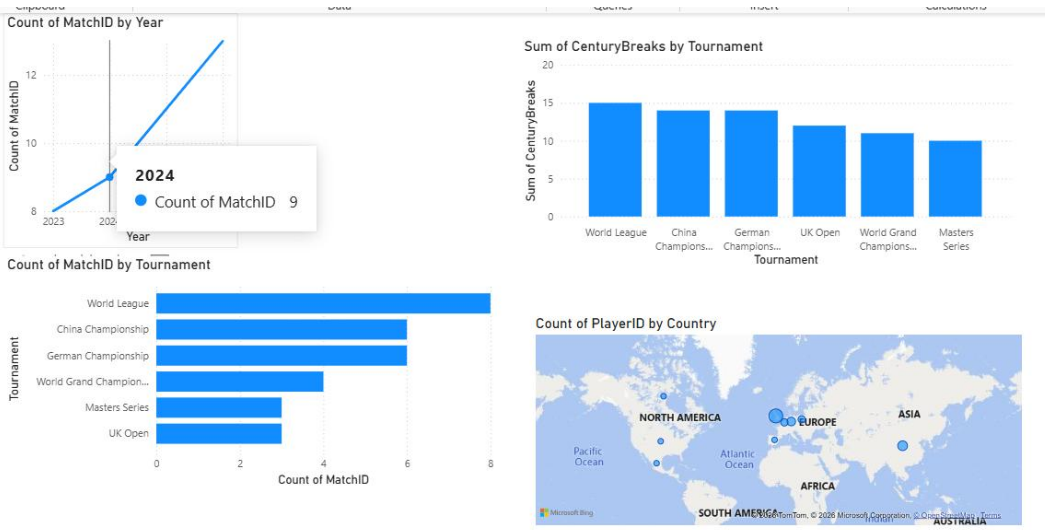
```
Matches Won =
CALCULATE(
    COUNTROWS(Matches),
    Matches[Winner] = SELECTEDVALUE(Players[PlayerName])
)
```

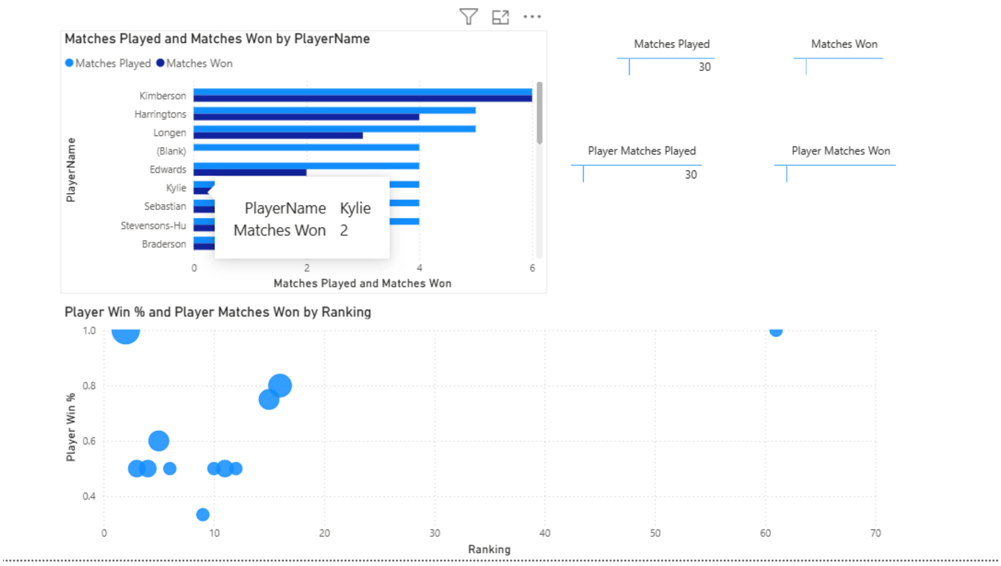
The third report shows matches played and matches won by player name. DAX measures are created for creating the Matches Player and Matches Won statistics.

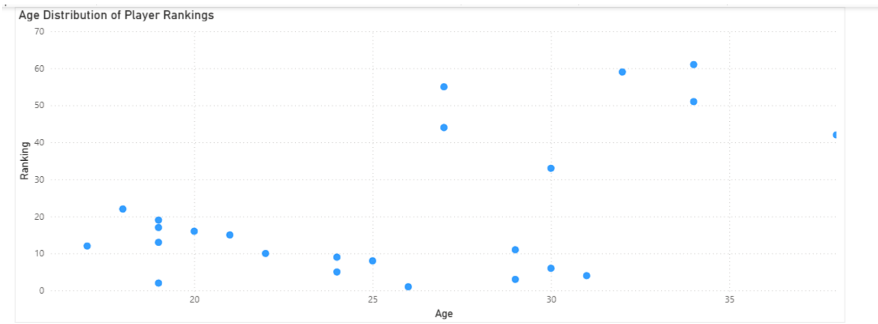
For Matches Played we use DISTINCTCOUNT to distinctly count all match ids in the matchplayers table.

For Matches Won we use CALCULATE to count all rows in the Matches table and use the Matches Winner column filtering out a certain players matches where they have won.

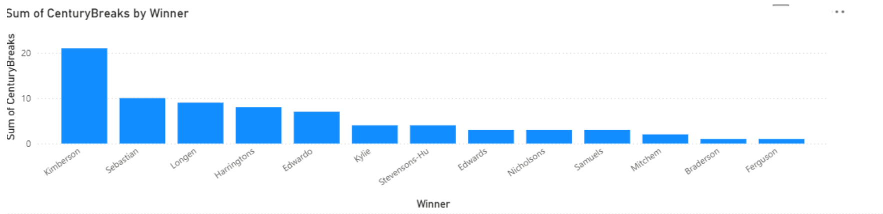
<u>All Reports With Example Screenshots</u>



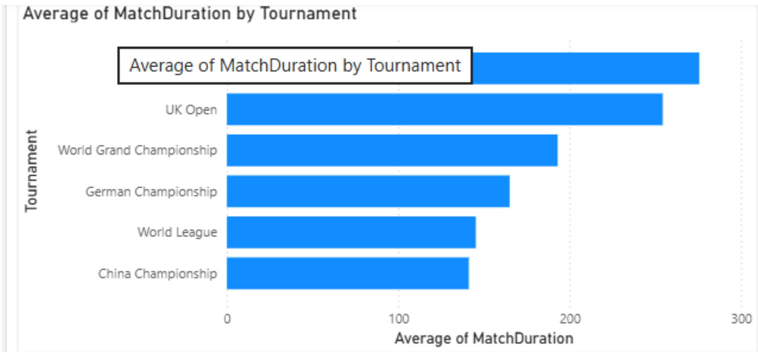




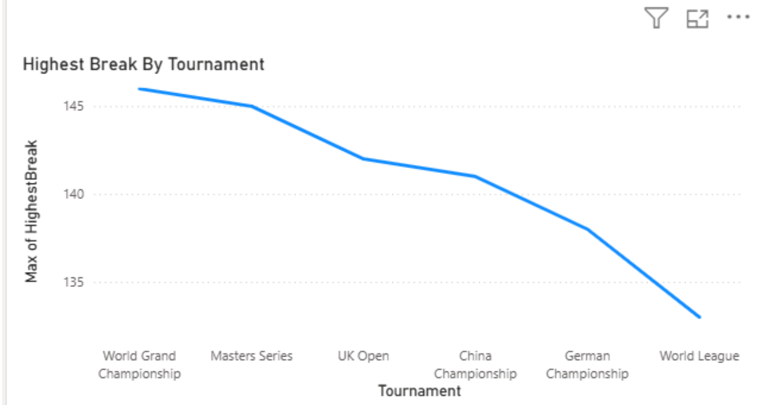
Age Distribution of Player Rankings
70
60
50
40
30
20
10
0
Ranking
20
25
30
35
Age



Sum of CenturyBreaks by Winner
Sum of CenturyBreaks
20
10
0
Kimberson
Sebastian
Longen
Harringtons
Edwardo
Kylie
Stevensons-Hu
Edwards
Nicholsons
Samuels
Mitchem
Braderson
Ferguson
Winner



Average of MatchDuration by Tournament
Average of MatchDuration by Tournament
UK Open
World Grand Championship
German Championship
World League
China Championship
Tournament
0
100
200
300
Average of MatchDuration



Highest Break By Tournament
145
140
135
Max of HighestBreak
World Grand Championship
Masters Series
UK Open
China Championship
German Championship
World League
Tournament

Tournament	Round of 32	Round of 16	Quarter Final	Semi Final	Final	Total
World League	145.00	135.33	176.00			**145.25**
World Grand Championship		171.00	176.00	254.00		**193.00**
UK Open			216.00	331.00		**254.33**
Masters Series				222.50	382.00	**275.67**
German Championship	166.00	136.50	192.50			**165.00**
China Championship	145.67	104.50		201.00		**141.17**
Total	**149.89**	**136.67**	**194.83**	**246.20**	**382.00**	**178.70**